Computer Architecture: An Introduction

Dr. Colin O’Flynn

Note on Slides, Notes, and Accommodations

- *Slides will generally be high-contrast where possible with this simple "PowerPoint Default" design to improve recognition if using assistance technology.*

- *Slides will be posted before classes as PDF (sometimes I forget – just ask in class if not there!).*

- *Lectures are recorded and posted in Brightspace – normally right after class, if not there by end of day check in with me.*

How Does a Computer Work?

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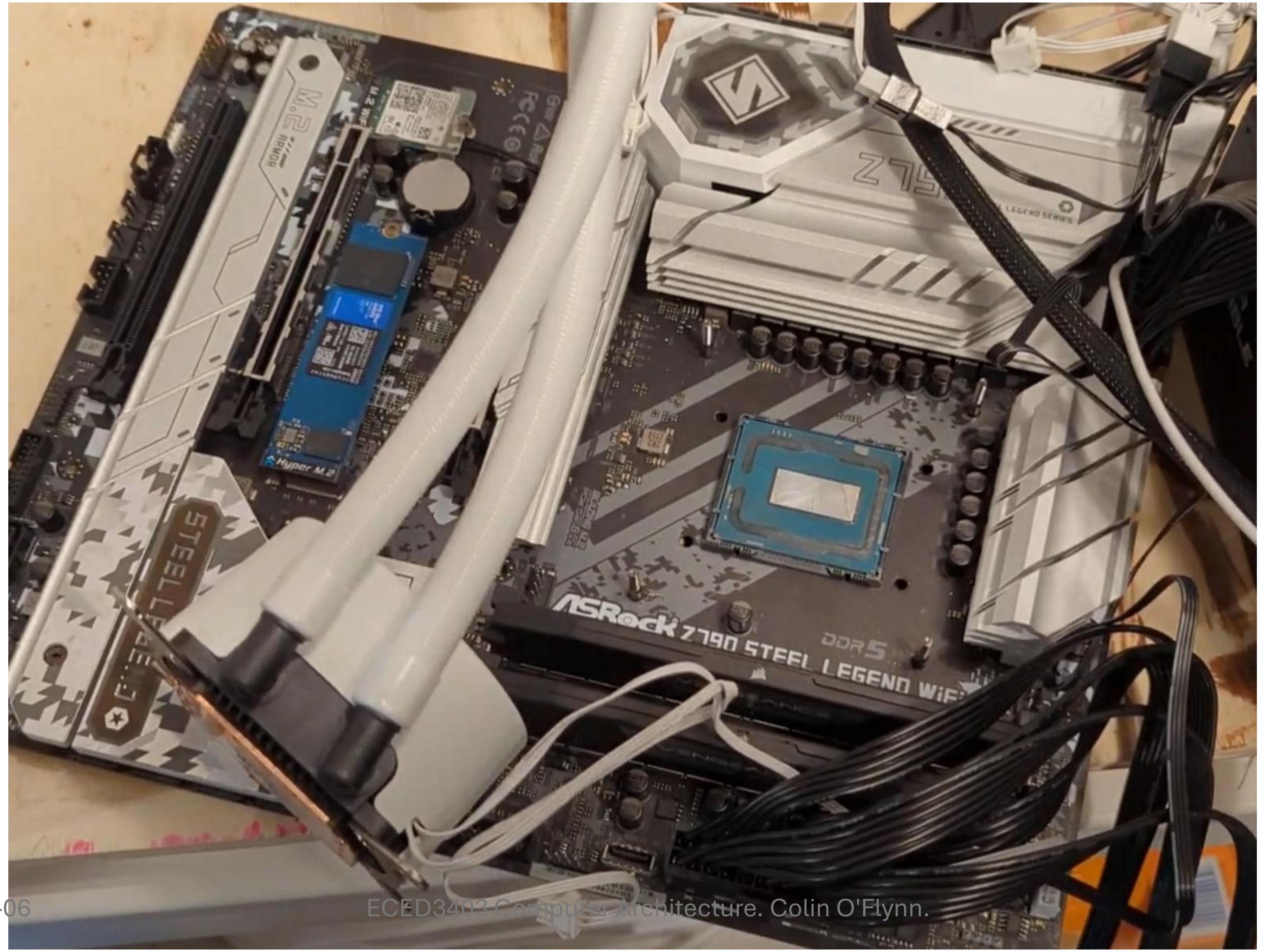

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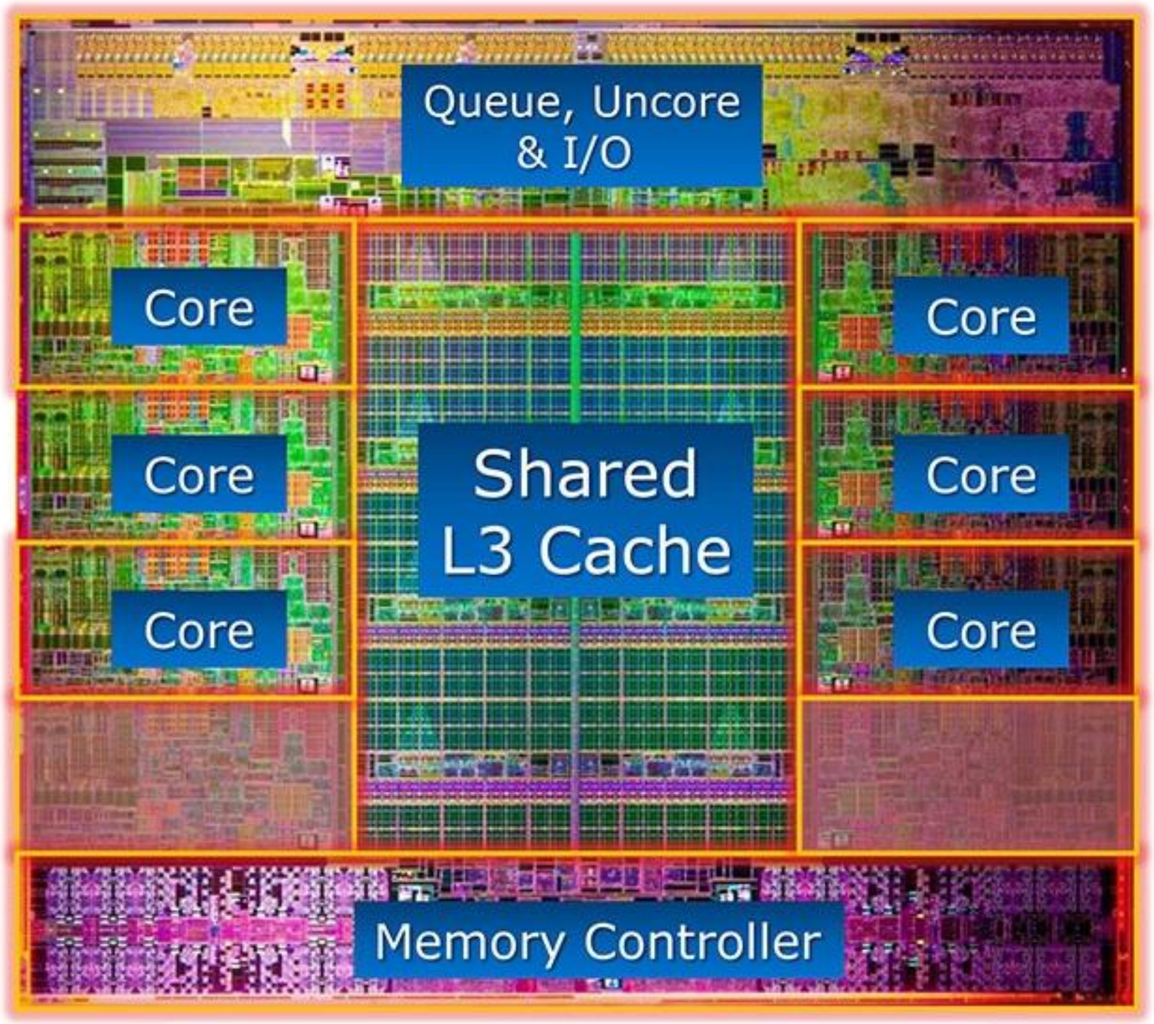
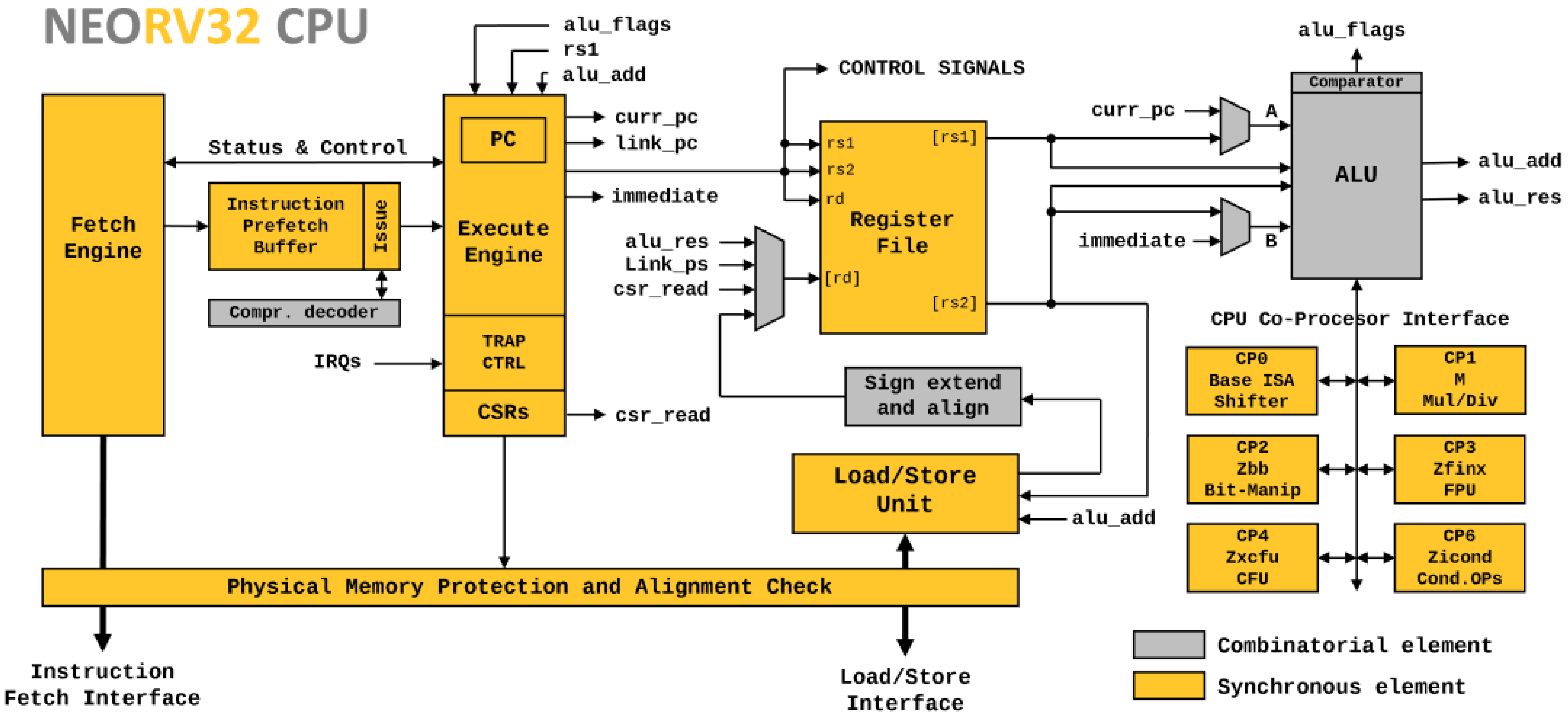
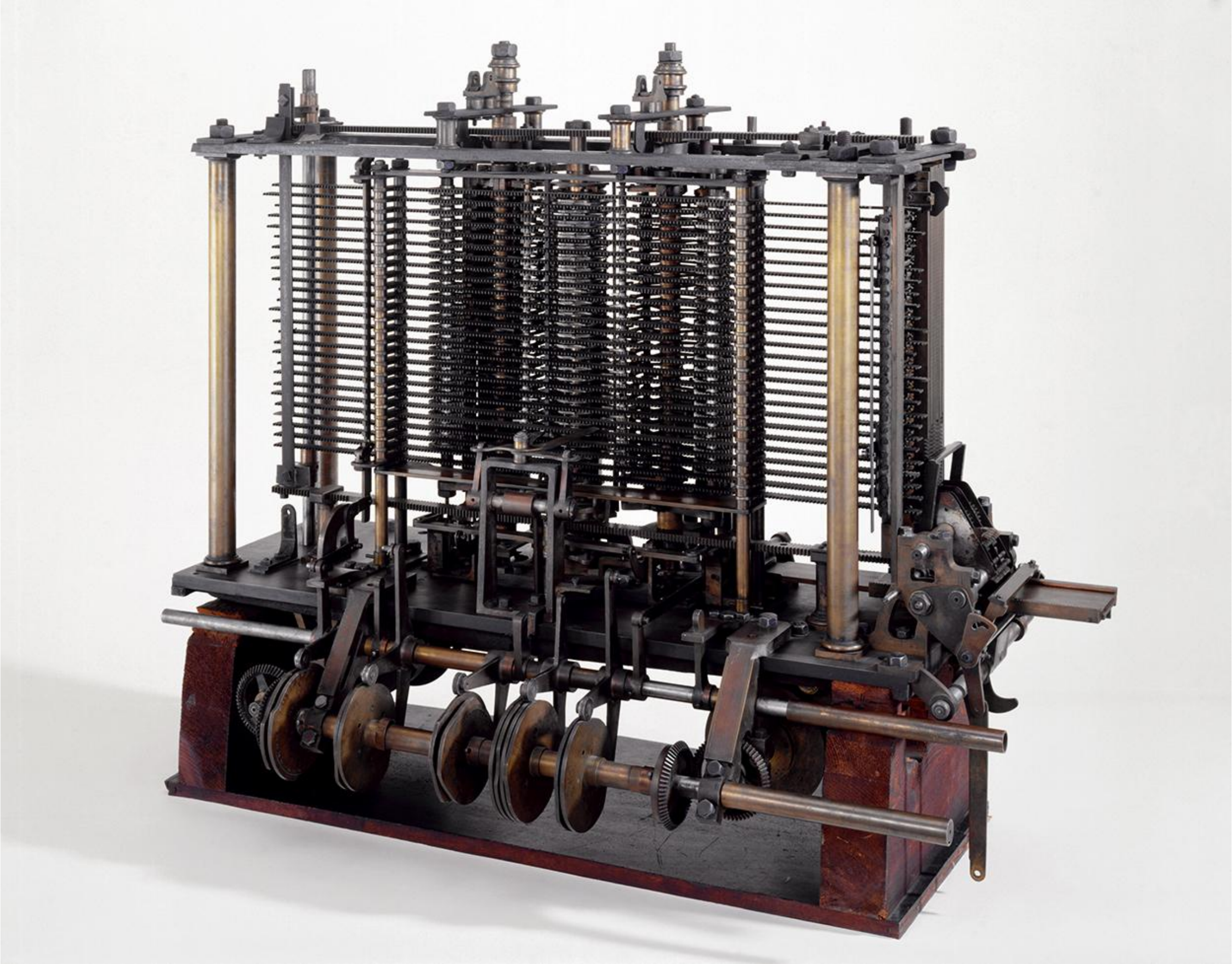


Intel® Core™ i7-3960X Processor Die Detail

How Does a Computer Work?



What Is A Computer?



Course & Topic Plans

- How Programs Run:
  - Compilers & Infrastructure
  - Linkers & Libraries
  - Assemblers
  - Machine Code
- Debuggers
- Performance Measurements
- Instruction Set Architectures
- Memory & Addressing
- The ALU & Flags
- The CPU Core
- Pipelines
- Exceptions
- Caches
- Peripherals & Busses
- Memory Safety
- Modern Architectures & Devices

Course Times / Dates

Lecture 1: Monday, 11:35-12:55AM, B229

Lecture 2: Wednesday, 2:35-3:55PM, B229

Lab/Tutorial: Tuesday 2:35-5:25PM, B229, Check Brightspace for schedule

Course Policies

- If Dalhousie is closed due to snow event I *will not* send a notice, there is no class. If we have many cancellations I may need to record missing lectures (there is some “flex” built in right now) or will use Tutorial times.
- If class will be cancelled for other reasons I will use Brightspace to announce this.
- Lab/Tutorials tend <u>not</u> to run every week – look for a Brightspace announcement to confirm if a lab is running (NO LAB next week).

Course Policies (From Syllabus)

Respectful Masking and Remote Attendance

If you suspect you are ill, I encourage you to use my class recordings to catch up and reduce the chance of spreading seasonal flu, COVID, or other illnesses that tend to make the rounds. I record classes and follow the absence reporting form (see below) to ensure students are not penalized for helping their fellow classmates in this way.

If you are recovering or "possibly" sick I also encourage respectful masking and you will likely see me doing so in case a family member or similar has been sick recently, as I don't want to risk causing a class outbreak myself!

Course Policies (From Syllabus)

On-Topic Discussion Policy

When attending course scheduled events such as lecture or labs, these times are to be used for computer architecture related learning and discussions. Students discussing other topics may be asked to leave the area if they wish to continue these discussions, as these topics may be distracting for other students. The course has very limited required attendance policy, so if you wish to discuss other topics you are not penalized for doing so elsewhere. Do not show up to the lectures if you wish to discuss other topics with your classmates.

Note this policy is in addition to the Dalhousie policies (such as student code of conduct, conduct in classroom, and diversity and inclusion policies), where even if the discussions is respectful I may still ask you to continue to elsewhere as it is off-topic for the course and distracts from the environment.

Course Policies (From Syllabus)

Absence Reporting (Illness or similar)

Students may use the self-reporting forms for missed assignments and labs at forms.engineering.dal.ca.

Students requesting academic accommodations must be registered with the Mark A. Hill accessibility center. Students with learning disabilities should register as early as possible and should identify themselves to the instructor early in the term and at least one week before any testing or activity which require accommodations. I cannot offer exemptions to rules without having an official accommodation registered – please do not email me personal information.

Any absence resulting in missed academic work must be reported using the Engineering Student Absence Reporting online system. This applies to both Student Declaration of Absence and Request for Accommodation. Visit forms.engineering.dal.ca for details and to submit a request.

Missed quizzes will be dropped. Missed assignments & labs will be given an extension.

How You Are Evaluated

- From the syllabus:
  - Assignments (4 per term) = 15%
  - Quizzes = 10%
  - Labs = 15%
  - Lab Exam = 20%
  - Final Exam = 40% **Final Exam Date TBD**

HINTS:

- The "Lab Exam" material closely follows the labs
- The "Final Exam" material closely follows the assignments

Quizzes

Quizzes

- Quizzes will be posted to Brightspace, and occur throughout the class. These are designed to give you rapid feedback on your class progress.
- You can drop the lowest two quiz marks, meaning your quiz mark will be made up of the remaining marks. This will automatically be applied to all students for your final grade calculation.
- There is no make-up for missed quizzes. The “free” dropped quizzes are designed to allow you to miss one or two. If you miss a quiz due to illness the mark will be dropped and your final mark consists of the remaining quiz average only.

Summary: If you do all the quizzes you should get close to 100% on this topic. Some will be trickier than others (there is a “free” one this week...).

Labs & Assignments

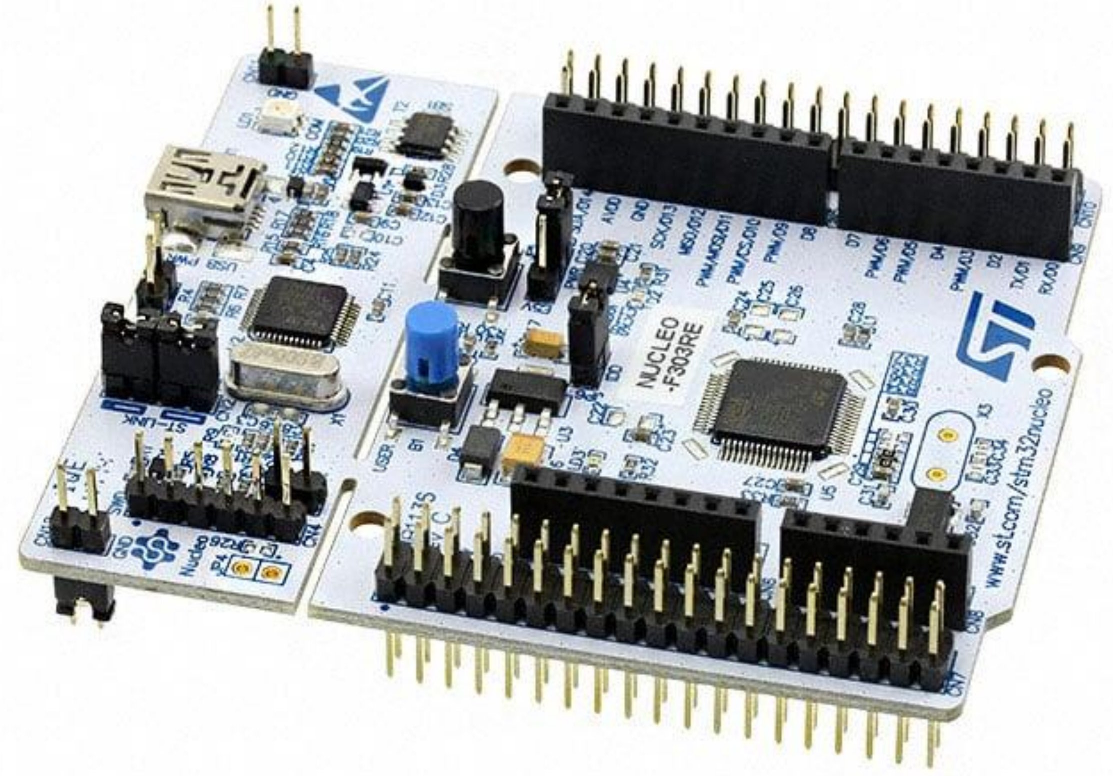
- There are four Labs & four Assignments in this course.
- They are in practice **combined** – each Lab/Assignment is around a specific topic.
- A practical aspect is done in a more group / interactive environment (the “Lab”).
- An assignment aspect is done individually and submitted as a normal assignment.
- All Labs will run on your laptop – you will be provided a development board.

Lab Topics

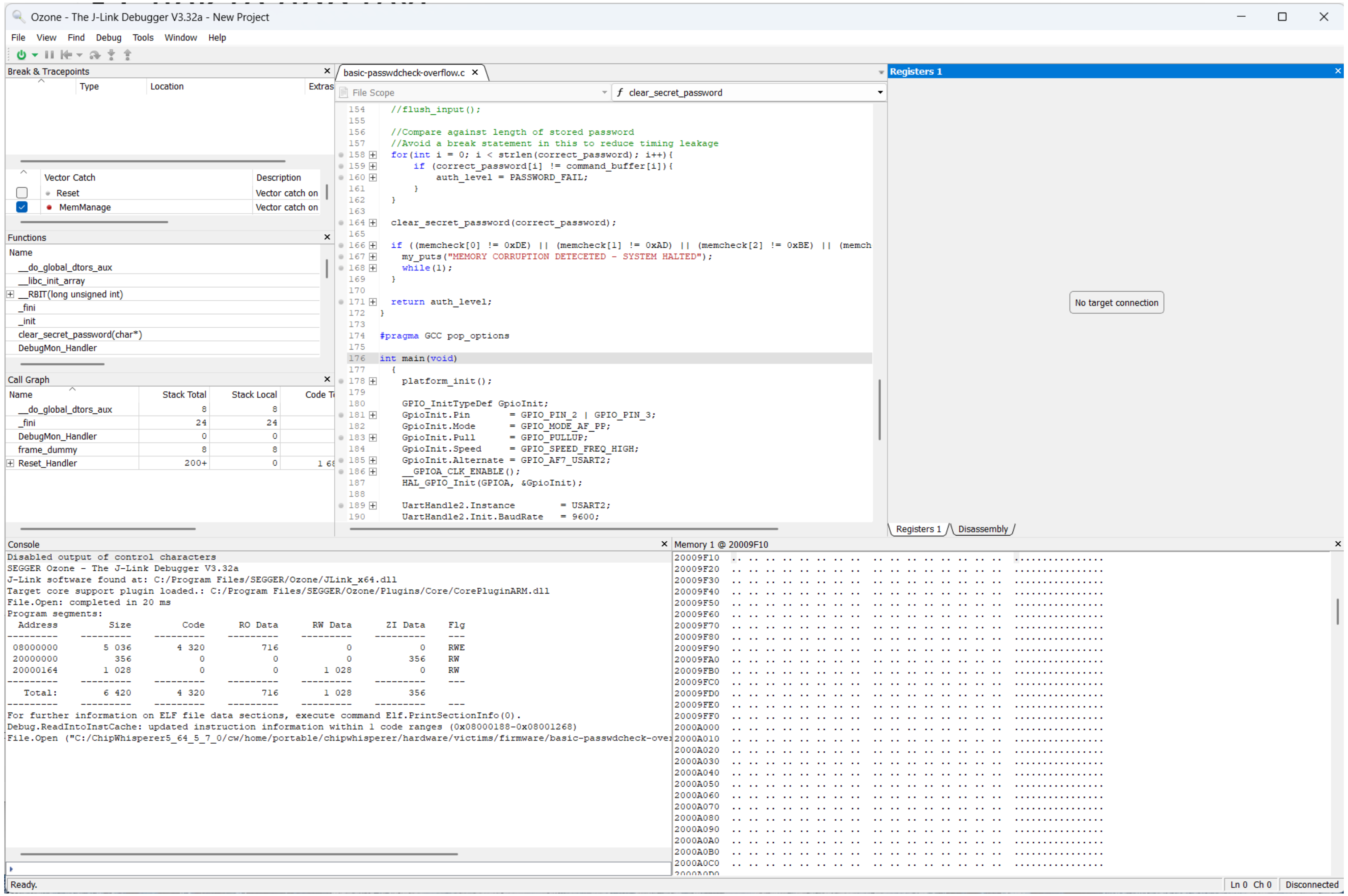
- Lab 1: Compiler Infrastructure
- Lab 2: Debuggers, Performance Evaluation
- Lab 3: Instruction Architecture, Microarchitecture, Caches, and Memory
- Lab 4: Interrupts & Exception Handling

(Subject to change)

STM32 Nucleo Development Board



Ozone Debugger



Ozone Debugger

<Live Demo If Working>

Notes on Quizzes, AI, and Final Exam(s)

- I follow Dalhousie's AI policy – work submitted should be your own (see syllabus for full statement).
- AI / LLM can be a <u>very useful</u> tool – it has limitations and I'll be showing you some of these as part of the class. Understanding <u>how</u> to use a tool and <u>when/where</u> to use it is a core part of engineering.
- I use the Final Exam as a validation of your personal knowledge – if you have completed assignments/labs yourself you'll find them easy, if using AI to complete assignments you'll find them difficult.

Homework: Quiz 1

- Please do “Quiz 1” ASAP.

Question 4 (0 points)

Listen

Have you used an embedded debugger (that lets you pause/stop code and inspect variables)?

- [ ] Yes - used in another class
- [ ] Yes - have used in my own projects or at work
- [ ] No
- [ ] Not sure what that is?

Question 5 (0 points)

Listen

Have you used any of the following types of devices? Select any that apply from courses, work, or personal projects.

- [ ] Embedded Linux on Raspberry Pi
- [ ] Arduino
- [ ] "Bare Bones" microcontroller
- [ ] RTOS on a microcontroller
- [ ] ARM Core Microcontroller
- [ ] AVR Core Microcontroller
- [ ] PIC Core Microcontroller
- [ ] 8051 Microcontroller
- [ ] RISC-V Microcontroller

Question 6 (0 points)

Listen

If asked to implement an interrupt/ISR on a microcontroller, I would:

- ( ) (Unfamiliar) Be unsure of how to proceed.
- ( ) (High-Level Knowledge) Need a SDK that provides an ISR stub.
- ( ) (Low-Level Knowledge) Be comfortable working with the interrupt table setup as well as any ISR needed.

Course Questions For Me

Questions or Concerns?

coflynn@dal.ca

If not we’ll pick this up on the first “normal” lecture day to introduce the first topic.